Module 7b slides: PK-PD (worked examples)

Module 7b - PK-PD Relationship · Lecture slides, worked examples · 12 Oct 2026

Name _____ Date _____

The minimum effective concentration (MEC or Ceff) in plasma for a certain antibiotic is 0.2 mcg/mL. The drug follows a one-compartment open model and has an apparent volume of distribution of 24 L and a first-order elimination rate constant of 0.15 h–1.

Slide 7 (Practice Problem)

What is the teff for a single 250-mg IV dose of this antibiotic?

Slide 7 (Practice Problem)

What is the new teff or for this drug if the dose were increased by a factor of four to 1000 mg?

Slide 7 (Practice Problem)

What is the new teff for this drug if the dose were increased by a factor of two to 500 mg?

Slide 8 What dose is required to double teff?